Khanh Nguyen

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EDUCATION

The University of Chicago — Chicago, IL
Master of Science in Computer Science - GPA: 3.73/4.0 — September 2025 - December 2026
Coursework: Parallel Programming, C Programming, Advanced Computer Architecture, High Performance Computing

Hanoi University of Science and Technology — Hanoi, Vietnam
Bachelor of Science in Artificial Intelligence and Data Science - GPA: 3.62/4.0 — August 2020 - August 2024
Coursework: Natural Language Processing, Big Data, Computer Vision, Operating Systems, Computer Architecture

PROFESSIONAL EXPERIENCE

Viettel AI — May 2023 - July 2025
Machine Learning Engineer
- Conducted research on speech recognition and implemented multiple state-of-the-art models from top-tier conferences, improved recognition accuracy by 22% for a smart TV platform with 2M+ users.
- Designed a domain-adaptation method for robot voice control that integrates Large Language Models (LLM)
- and text-to-speech models, increasing intent-detection accuracy by 15%.
- Conducted large-scale distributed ASR training on 50k hours of audio, increasing training speed by 3x by optimizing the data loading process with bucketing datasets.
- Developed an end-to-end automated pipeline for crawling, preprocessing, and refining code-switching audio datasets from YouTube, improving movie-query recognition accuracy by 35%.
- Implemented and trained a multimodal ASR model that can bias transcription using a user-provided prompt.
- Developed a multimodal Speech-LLM by integrating a Conformer encoder with a Qwen-2.5 backbone via LoRA fine-tuning to enable parameter-efficient, high-fidelity acoustic reasoning and speech understanding.

Vbee.ai — June 2022 – February 2023
AI Intern
- Developed an audio call classification system with XGBoost, ONNX, and FastAPI, achieving 87% accuracy.
- Designed, implemented, and systematically benchmarked various end-to-end Spoken Language Understanding architectures, providing data-driven insights for model selection and optimization.

PROJECT EXPERIENCE

High Performance Neural Network Implementation (C++/CUDA) — January 2026 – February 2026
- Built a feedforward neural network from scratch for MNIST classification, implementing forward/backward propagation, mini-batch SGD, and cross-entropy loss.
- Engineered custom CUDA kernels for matrix multiplication operations, achieving 90% of cuBLAS throughput on an NVIDIA V100 GPU.
- Maximized GPU utilization by masking data-transfer latency, utilizing pinned memory, and multiple asynchronous CUDA streams to overlap host-to-device loading with computation.

Cache Simulator (C++) — December 2025 – January 2026
- Developed a configurable CPU and two-level cache (L1/L2) emulator capable of modeling memory hierarchy and executing custom pseudo-assembly programs.
- Implemented diverse hardware simulation features, including various cache mapping strategies (direct, n-way, fully associative), replacement policies (LRU, FIFO, Random), and benchmarking algorithms like matrix multiplication.
- Conducted quantitative performance analysis by profiling memory events (hits/misses) to evaluate the architectural impact of cache capacity, block size, associativity, and cache thrashing on overall system efficiency

Ezspeech (Pytorch) — December 2024 – Present
- Developed a modern, easy-to-use speech recognition toolkit using PyTorch Lightning; provided modern ASR models with clean APIs for training, evaluation, and deployment.
- Enabled easy loading of NVIDIA NeMo models for fine-tuning and customization.

SKILLS

Programming Languages: Python, Java, C, C++, OpenMP, Go, CUDA, MPI
Technologies/Frameworks: PyTorch, Docker, FastAPI, WebSockets, Vector Databases, Triton Inference Server, NVIDIA NeMo, HuggingFace